



SAP Commerce Cloud

Cloud Readiness Check - City of Belgrade

February 8, 2023

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Document Revision History

Date	Revision	Author	Summary of Changes
February 1, 2023	1.0	Stephan Beutel	Initial Version

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1 INTRODUCTION

This report contains the findings from the Cloud Readiness Check (CRC), which aims to reduce risks when deploying your solution for the first time on SAP Commerce Cloud. A review of your solution has been executed in order to highlight potential issues and make recommendations on areas of your solution that may experience issues on or after go-live.

The CRC is intended as a check to ensure your solution is reviewed for many of the issues commonly found prior to going live in your production environment. It does **not** include system nor application fixes as a result of the review performed. Any actions recommended as a result of the CRC, will have to be reviewed and completed by City of Belgrade and/or SAP CX.

If an in-depth review is required, we encourage you to further explore how review and validation services offered by SAP can help to further boost the quality of your project. Please contact your Customer Engagement Manager for more details.

1.1.1 Activities

The following tasks were planned and executed as of February 1, 2023 and presented to Customer. This review is based on the latest stable code revision identified by n/a.

Area	Description
Project Setup	Review of your configuration to ensure it meets the standards for deploying on Commerce Cloud
Source Code Analysis	A high-level static source code analysis of the custom codebase using SonarQube; Type system validation;
Integrations	Brief inspection of your integrations for compatibility and meets recommended design patterns. Running/testing of the integrations is out of scope.
Performance Testing Approach	A high-level evaluation to look for possible performance problems that will require further investigation before go-live
Data Maintenance	Review of the data being generated: what data is live, what data is being archived, and what data can be removed to keep the system running at peak performance
Application Security Review	Review of security practices. Ensuring best practices are being followed to guarantee a secure system

Each chapter in this report contains detailed information on improvements, further issues and recommendations to be investigated in each area. These activities are in line with the recommended practices collected in our methodology. Further information can be found in the [official documentation](#) or on <https://www.sap.com/cxworks/>.

2 SUMMARY OF RECOMMENDATIONS

The following table provides a list of all recommendations found during the execution of the Cloud Readiness Check. Each recommendation has a reference to a corresponding numbered section with the full details.

- **Blocker** – Needs to be fixed before go-live. These are very likely to result in a negative impact on your solution at go-live.
- **High** – Should be fixed by go-live. If not possible, plans should be included to fix as part of fast-follow release as there is still high chance of an issue occurring.
- **Medium** – Does not need to be fixed prior to go-live. Should create plans to fix within 3-5 months of going live.
- **Low** - Does not need to be fixed prior to go-live. Should create plans to fix within 3-12 months of going live, as it may have a detrimental impact the longer your solution runs.

Issue	Recommendation	Section	Priority
Commerce Version	Your current version 2105 of SAP Commerce . The EOMM is August 19, 2023 Upgrade to 2211 recommended	3.1	BLOCKER
Endpoint Configuration	Unused endpoints can be removed and resources moved to active endpoints	3.3	BLOCKER
Performance Testing	A high-level evaluation of the performance testing approach to ensure that the application is being properly profiled. See the section for more details.	6.2	BLOCKER
Environment Sizing	We have provided some recommendations on how the aspects of your solution could be scaled up/down proactively. See section for detailed recommendations	6.3	BLOCKER
Unencrypted type attributes	Transparent Attribute Encryption is not enabled. Sensitive data may have been stored on the database unencrypted.	8	BLOCKER
Missing password encoding	Some passwords were found to not be encoded. Review password encoding configuration.	8	BLOCKER
Manifest file	The following sections of your manifest file were found to have issues: • commerceSuiteVersion See section 3.2 for detailed recommendations.	3.2	HIGH
Bugs & Vulnerabilities	Address 4 Blocker bugs found by code analysis before go-live	4.1	HIGH
Type System	We identified some issues with custom types you have defined for your solution. Please see the section for individual issues identified	4.3	HIGH
Critical Business Processes	An analysis of your cronjobs identified some issues with how they are defined. See section for detailed list.	4.6	HIGH
Incompatible Integrations	Integrations are following the recommended approach.	5.2	HIGH
Test Data	We could not find any indication of realistic test data, which is critical to helping prove viability of your SAP Commerce Cloud solution.	6.1	HIGH
Large Tables	Your solution contains some unexpectedly large tables, which is usually an indication that cleanup jobs aren't in place. Consider cleaning up the following:	7.1	HIGH

	JobLogs SavedValues SavedValueEntry Props ProcessTaskLog TaskLogs Cart CartEntry		
CronJob Cleanup	Configure and activate CleanupCronJobStrategy based on your audit requirements to ensure you are clearing out cronjobs in order to not fill up your database.	7.2	HIGH
Promotions	No promotions used	7.6	HIGH
Data Migration	Data will be copied to CCV2 by using impex files.	7.7	HIGH
Large number of users granted access to Cloud Portal	There are many users with the CUSTOMER SYS ADMIN role, which allows a person to add new users. Limit access to this role. There are many users defined with access to the Cloud Portal. This should be treated as an admin tool and access should be limited only to those administering the Commerce Cloud solution	8	HIGH
SOLR Indexing strategy	Validate if it is following the recommendation of using Two-Phase strategy. Indexes with "DIRECT" mode: -	3.1	MED
Overridden Properties	You have defined properties that will be overridden by Commerce Cloud. Review the default configurations and determine if other properties should be used	3.1	MED
Redirects	No redirection sets were defined. We commonly see redirects for a Commerce Cloud solution. They can be defined for an endpoint, enabling vanity & marketing URLs and site migrations. Confirm it is ok to not have any redirect sets defined. See documentation	3.3	MED
Static Artifacts	No static artifacts were found. They can be uploaded for an endpoint, enabling search engines to better understand the site. Confirm it is ok to not have any static artifacts defined. See documentation	3.3	MED
VPN	A virtual private network has not been found. These are typically used to access Commerce Cloud from a private network or privately hosted system. Confirm it is ok to not have any VPNs defined. See documentation	3.4	MED
Coverage	Your code coverage was 35.4%. SAP recommends to focus on coverage of 80% in the Facades and Service Layers with Integration Tests. See attached report for details on your test coverage	4.1	MED
Search Restrictions	There are some custom search restrictions found that do not follow recommended practices. See section for detailed list.	4.5	MED
Integration Recommendations	A high-level review of your integrations identified some potential issues. See section for considerations.	Error! Reference source not found.	MED
Sample Data	Sample test data is critical for test environments, but it shouldn't exist in Production environments.	7.3	MED

Update JSON

It's recommended to use a deployment/update JSON to prevent Sample data from being imported during platform update.

7.4

LOW

3 PROJECT SETUP

3.1 Compatibility & Startup

Property	Further Reading	Issues Found
The current SAP Commerce version is supported by Commerce Cloud. It is recommended to be on the latest version of SAP Commerce in order to be able to take advantage of all the features.	Supported Releases and Application Version	Yes
The current SAP Commerce SOLR setup is supported by SAP Commerce Cloud.	Third Party Compatibility	
The following properties are configured in your source control but will be overridden by Commerce Cloud: addonfilter.active - /opt/hybris/config/local.properties; java.mem - /opt/hybris/config/local.properties; default.session.timeout - /opt/hybris/config/local.properties	Managed Properties	Yes
The following extensions are in use but have been marked for deprecation: No deprecated extensions were found	Deprecation Status	No

3.2 Manifest File

The manifest is a key component in telling the build automation tooling in Commerce Cloud how to build your solution. We have reviewed your manifest files and can provide the following recommendations:

Property	Recommendation	Issues Found
commerceSuiteVersion	SAP recommends to not specify the patch version in order to always have the latest security patches applied with each build. This may lead to inconsistent testing between your local development and Cloud environments when new patches are released without noticing. If the manifest is specifying a fixed patch version, it must make sure that project team is manually keeping track of patches. Currently version 2105 is used, however the latest version is 2211. It is recommended you update to the latest version.	Yes EOMM Aug 19, 2023
extensions	Configuration Reuse should be used to have consistency between local development and cloud environments	No
properties	It is recommended that you do not include properties in your manifest file. These should be defined in the correct *.properties file and loaded from source as part of configuration reuse	No
aspects	For each aspect listed below the webapp was found to be defined incorrectly. Consider moving to the suggested aspect: No issues found	No

See [Commerce manifest components](#) and [manifest examples](#).

3.3 Cloud Portal Configuration

A review of Cloud Portal was executed, including a review of the configurations of your endpoints. The following issues were identified:

Issue	Description
Unused Endpoints	Storefront, JS Storefront
Incorrect caching configuration	-
IP Filters not defined	-
Certificate Issues	No custom SSL certificate in place.
Maintenance Pages	-
Redirects	-
Static Artifacts	-

See [Cloud Portal](#) for more information.

3.4 VPN & NAT Configuration

A virtual private network **has not been found**. These are typically used to access Commerce Cloud from a private network or privately hosted system. Confirm it is ok to not have any VPNs defined. See [documentation](#)

4 SOURCE CODE ANALYSIS

4.1 Quality Gates

An analysis of your custom code was executed in Sonar using the rulesets provided in the SAP Commerce Core release (e.g. java-hybris-profile.xml). We have attached the full report. For more, see [Measuring Code Quality with Sonar](#).

4

Bugs

Reliability

E

0

Vulnerabilities

Security

A

2

Security Hotspots

0.0%

Reviewed

Security Review

E

25d

Debt

748

Code Smells

Maintainability

A



35.4%

Coverage on 7.5k Lines to cover

174

Unit Tests



4.6%

Duplications on 23k Lines

117

Duplicated Blocks

Gate	Recommendation	Issues Found
Bugs & Vulnerabilities	Address Blocker bugs before go-live and may consider criticals later also. Blockers: 4 Criticals: 20	Yes
Duplication	Code duplication above 5% is not recommended. Backlog tickets should be created to deal with this. Duplication: 4.6%	No
Technical Debt	Technical debt above 30 days at this point in the project could indicate underlying issues that may cause problems down the line. Backlog tickets should be created to deal with this debt as part of your development cycle. It should be monitored to ensure it's not increasing. Technical Debt: 25 days	No
Coverage	Most source code should be automatically tested. Custom classes that are not under test may run the risk of containing bugs or potentially cause increased maintenance of the project in the future. SAP recommends to focus on coverage (80%) in the Facades and Service Layers with Integration Tests. See attached report for details on your test coverage. Coverage: 35.4%	Yes

4.2 OWASP Security Standards

Sonar has identified the following potential security issues:

Issue	Recommendation
n/a	

4.3 Type System Validation

The Type System Validation (TSV) Tool is a tool created by SAP Services to evaluate custom types in your solution. The following items in the type system must be investigated:

ELEMENT	DESCRIPTION	FILENAME	LINE #
Address.phoneNumbers	Collections have to be only defined for dynamic and jalo attributes	/custom/bgh/bghcore/resources/bghcore-items.xml	216
BghPaginatedCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.	/custom/bgh/bghcore/resources/bghcore-items.xml	284
BghTtsGenerationCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.	/custom/bgh/bghcore/resources/bghcore-items.xml	296
Category.sharingDescription	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,hsqldb,sap).	/custom/bgh/bghcore/resources/bghcore-items.xml	178
Category.summary	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,hsqldb,sap).	/custom/bgh/bghcore/resources/bghcore-items.xml	165
Product.additionalInformation	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,hsqldb,sap).	/custom/bgh/bghcore/resources/bghcore-items.xml	100
Product.keyInformation	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,hsqldb,sap).	/custom/bgh/bghcore/resources/bghcore-items.xml	113
Product.stepsDescription	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,hsqldb,sap).	/custom/bgh/bghcore/resources/bghcore-items.xml	87
Product.title	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,hsqldb,sap).	/custom/bgh/bghcore/resources/bghcore-items.xml	63
PsServiceProduct2BghDocMediaRelation	Any side of a relation that has cardinality='many' should have collectiontype='set' unless absolutely necessary	/custom/bgh/bghcore/resources/bghcore-items.xml	31

4.4 Database Compatibility

Some attributes in your data model have been declared with database specific mapping that may not be correct for Microsoft Azure SQL database used by Commerce Cloud. In the help page [Specifying a Deployment for](#)

[Commerce Platform Types](#) under section 'Advanced Deployment' you will find further information's and the standard mapping of HYBRIS.LONG_STRING to the dedicated RDBMS data types and the hint to core-advanced-deployment.xml in the 'Advanced Deployment' section.

The following attributes should be reviewed to ensure they are mapped correctly (see table below for reference):

Data Type	Database Mapping			
	sap	hsqldb	mysql	sqlserver
java.lang.String	nvarchar(255)	VARCHAR(255)	varchar(255)	nvarchar(255)
String	nvarchar(255)	VARCHAR(255)	varchar(255)	nvarchar(255)
java.lang.Float	decimal(20,5)	float	float(20,5)	float
java.lang.Double	decimal(30,8)	double	double	float
java.lang.Byte	smallint	smallint	smallint	integer
java.lang.Character	smallint	smallint	smallint	char(4)
java.lang.Short	integer	smallint	integer	integer
java.lang.Boolean	decimal(1,0)	tinyint	tinyint(1)	tinyint
java.lang.Long	bigint	bigint	bigint	bigint
java.lang.Integer	bigint	int	integer	integer
float	decimal(20,5) DEFAULT 0	float default 0	float(20,5) DEFAULT 0	float default 0
double	decimal(30,8) DEFAULT 0	double default 0	double DEFAULT 0	float default 0
byte	integer DEFAULT 0	smallint default 0	smallint DEFAULT 0	integer default 0
char	integer DEFAULT "	smallint default 0	smallint DEFAULT 0	char(4) default "
short	integer DEFAULT 0	smallint default 0	integer DEFAULT 0	integer default 0
boolean	decimal(1,0) DEFAULT 0	tinyint default 0	tinyint(1) DEFAULT 0	tinyint default 0
long	bigint DEFAULT 0	bigint default 0	bigint DEFAULT 0	integer default 0
int	bigint DEFAULT 0	int default 0	integer DEFAULT 0	integer default 0
java.util.Date	timestamp	timestamp	datetime	datetime2
java.math.BigDecimal	decimal(30,8)	decimal(30,8)	decimal(30,8)	decimal(30,8)
java.io.Serializable	blob	longvarbinary	longblob	image
HYBRIS.LONG_STRING	nvarchar(5000)	longvarchar	text	nvarchar(max)
HYBRIS.JSON	nclob	longvarchar	longtext	nvarchar(max)
HYBRIS.COMMA_SEPARATED_PK	nvarchar(5000)	longvarchar	text	nvarchar(max)
HYBRIS.PK	bigint	BIGINT	bigint	bigint

4.5 Search Restriction

All active search restrictions were reviewed. The following search restrictions were reviewed to ensure they follow recommended practices

No issues found.

Restriction	Restricted Type	Principal	Query
Frontend_Category	Category	customergroup	{item:catalogVersion} IN (?session.catalogversions)
Frontend_ProductOfflineDate	Product	customergroup	({onlineDate} IS NULL OR {onlineDate} <= ?session.user.currentDate) AND ({offlineDate} IS NULL OR {offlineDate} >= ?session.user.currentDate)
Frontend_ProductApprovalStatus	Product	customergroup	{approvalStatus} = 8796100886619
Frontend_Product	Product	customergroup	{item:catalogVersion} IN (?session.catalogversions)
Frontend_Keyword	Keyword	customergroup	{item:catalogVersion} IN (?session.catalogversions)
Frontend_ProductReference	ProductReference	customergroup	EXISTS (({SELECT {pk} FROM {Product} WHERE {pk}={item:target} })))
Backend_visibility	CMSItem	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL
Backend_visibility	PriceRow	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL
Backend_visibility	DisableProductBundleRule	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL
Backend_visibility	AbstractAsSearchConfiguration	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL
Backend_visibility	BundleTemplate	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL
Backend_visibility	AttributeValueAssignment	employeeegroup	{systemVersion} IN (?session.catalogversions) OR {systemVersion} IS NULL
Backend_visibility	ClassificationAttributeUnit	employeeegroup	{systemVersion} IN (?session.catalogversions) OR {systemVersion} IS NULL
Backend_visibility	ClassificationAttribute	employeeegroup	{systemVersion} IN (?session.catalogversions) OR {systemVersion} IS NULL
Backend_visibility	TaxRow	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL
Backend_visibility	Keyword	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL

Restriction	Restricted Type	Principal	Query
Backend_visibility	BundleSelectionCriteria	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL
Backend_visibility	PSConsentDeclaration	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL
Backend_visibility	AbstractAsSearchProfile	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL
Backend_visibility	AbstractConfiguratorSetting	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL
Backend_visibility	AbstractAsItemConfiguration	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL
Backend_visibility	DiscountRow	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL
Backend_visibility	BundleTemplateStatus	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL
Backend_visibility	AsSearchProfileActivationSet	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL
Backend_visibility	ClassAttributeAssignment	employeeegroup	{systemVersion} IN (?session.catalogversions) OR {systemVersion} IS NULL
Backend_visibility	ChangeProductPriceBundleRule	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL
Backend_visibility	ClassificationAttributeValue	employeeegroup	{systemVersion} IN (?session.catalogversions) OR {systemVersion} IS NULL
Backend_visibility	PSProductInfo	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL
Backend_visibility	MediaContainer	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL
Backend_visibility	CMSRelation	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR {catalogVersion} IS NULL
Backend_visibility	Product	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR EXISTS ({{SELECT {pk} FROM {Category AS cat JOIN CategoryProductRelation AS c2p ON {cat:pk}={c2p:source} } WHERE {c2p:target}={item:pk} AND {cat:catalogVersion} NOT IN({{SELECT {pk} FROM { ClassificationSystemVersion} }}})}})

Restriction	Restricted Type	Principal	Query
Backend_visibility	VariantProduct	employeeegroup	{catalogVersion} IN (?session.catalogversions) OR EXISTS ({{SELECT {pk} FROM {Product AS base} WHERE {base:pk}={item:baseProduct} }})
Backend_visibility	Catalog	employeeegroup	{pk} IN ({{SELECT {catalog} FROM {CatalogVersion} }}) OR NOT EXISTS ({{SELECT {pk} FROM {CatalogVersion* AS cv} WHERE {cv:catalog}={item:pk}}})
Backend_visibility	CatalogVersion	employeeegroup	{pk} IN (?session.catalogversions)
Frontend_Group_Price Rows	PriceRow	customergroup	({item:product} IS NOT NULL OR{item:catalogVersion} IS NULL OR {item:catalogVersion} IN (?session.catalogversions))
Frontend_Group_TaxRows	TaxRow	customergroup	({item:product} IS NOT NULL OR{item:catalogVersion} IS NULL OR {item:catalogVersion} IN (?session.catalogversions))
Frontend_Group_DiscountRows	DiscountRow	customergroup	({item:product} IS NOT NULL OR{item:catalogVersion} IS NULL OR {item:catalogVersion} IN (?session.catalogversions))
Backoffice_Media_Catalog_Version	CatalogVersion	employeeegroup	{catalog} not in ({{SELECT {pk} FROM {Catalog*} where {id}='_boconfig'}})
Backoffice_Media_Catalog	Catalog	employeeegroup	{id} not in ('_boconfig')
Backoffice_Configuration_Media	Media	employeeegroup	{code} not in ('cockpitng-config', 'cockpitng-widgets-config')
CustomerReviewRestrictions	CustomerReview	customergroup	{blocked} = 0 OR {blocked} IS NULL
webhookServiceVisibility	IntegrationObject	integrationusergroup	((code)='WebhookService' AND EXISTS ({ select {ug:PK} from {UserGroup as ug} where {ug:PK} IN (?session.user.groups) and {ug:uid} = 'webhookservicegroup' } })) OR ({code}!='WebhookService')
outboundChannelConfigVisibility	IntegrationObject	integrationusergroup	((code)='OutboundChannelConfig' AND EXISTS ({ select {ug:PK} from {UserGroup as ug} where {ug:PK} IN (?session.user.groups) and {ug:uid} = 'outboundsyncgroup' } })) OR ({code}!='OutboundChannelConfig')
outboundMonitoringIntegrationVisibility	IntegrationObject	integrationusergroup	((code)='OutboundIntegrationMonitoring' AND EXISTS ({ select {ug:PK} from {UserGroup as ug} where {ug:PK} IN (?session.user.groups)

Restriction	Restricted Type	Principal	Query
			and {ug:uid} = 'integrationmonitoringgroup' } })) OR ({code}!='OutboundIntegrationMonitoring')
runtimeAttributeVisibility	IntegrationObject	integrationusergroup	(({code}='RuntimeAttributeService' AND EXISTS ({{ select {ug:PK} from {UserGroup as ug} where {ug:PK} IN (?session.user.groups) and {ug:uid} = 'integrationservicegroup' } })) OR ({code}!='RuntimeAttributeService')
scriptServiceVisibility	IntegrationObject	integrationusergroup	(({code}='ScriptService' AND EXISTS ({{ select {ug:PK} from {UserGroup as ug} where {ug:PK} IN (?session.user.groups) and {ug:uid} = 'scriptservicegroup' } })) OR ({code}!='ScriptService')
integrationServiceVisibility	IntegrationObject	integrationusergroup	(({code}='IntegrationService' AND EXISTS ({{ select {ug:PK} from {UserGroup as ug} where {ug:PK} IN (?session.user.groups) and {ug:uid} = 'integrationservicegroup' } })) OR ({code}!='IntegrationService')
inboundMonitoringIntegrationVisibility	IntegrationObject	integrationusergroup	(({code}='InboundIntegrationMonitoring' AND EXISTS ({{ select {ug:PK} from {UserGroup as ug} where {ug:PK} IN (?session.user.groups) and {ug:uid} = 'integrationmonitoringgroup' } })) OR ({code}!='InboundIntegrationMonitoring')
InboundConsentTemplateIntegrationVisibility	IntegrationObject	integrationusergroup	(({code}='InboundConsentTemplate' AND EXISTS ({{ select {ug:PK} from {UserGroup as ug} where {ug:PK} IN (?session.user.groups) and {ug:uid} = 'InboundConsentTemplateGroup' } })) OR ({code}!='InboundConsentTemplate')
YFormDefinition_Latest_Versions	YFormDefinition	yformsmanagergroup	{item:latest} = 1
YFormDefinition_Enabled_Forms	YFormDefinition	yformsmanagergroup	{item:status} IN ({{SELECT {s.pk} FROM {YFormDefinitionStatusEnum as s} WHERE {s.code}='ENABLED' }})
inStoreBundleVisibility	BundleTemplate	customergroup	{item:catalogVersion} IN (?session.catalogversions)
FrontEnd_Reviews	CustomerReview	customergroup	{approvalStatus}=8796107472987
Sync_Only_Approved_Pages_Restriction	AbstractPage	cmssyncuser	{item:approvalStatus}=8796106522715
Sync_Only_Approved_Slots_For_Page_Restriction	ContentSlotForPage	cmssyncuser	EXISTS ({{ SELECT 1 FROM { ContentSlotForPage AS slot4page JOIN AbstractPage AS p ON {p:pk} = {slot4page:page} } WHERE {item:pk} = {slot4page:pk} }})

Restriction	Restricted Type	Principal	Query
Sync_Only_Approved_Slots_Restriction	ContentSlot	cmssyncuser	({item:synchronizationBlocked} IS NULL OR {item:synchronizationBlocked} = 0)
Sync_Only_Approved_Components_Restriction	AbstractCMSComponent	cmssyncuser	({item:synchronizationBlocked} IS NULL OR {item:synchronizationBlocked} = 0)
Sync_Only_Approved_Restrictions_Restriction	AbstractRestriction	cmssyncuser	({item:synchronizationBlocked} IS NULL OR {item:synchronizationBlocked} = 0)
Sync_Only_Approved_Media_Container_Restriction	MediaContainer	cmssyncuser	{item:pk} IN ({{ SELECT {mediaComponent:media[ANY]} FROM {AbstractMediaContainerComponent AS mediaComponent} }}
Sync_Only_Approved_Media_Restriction	Media	cmssyncuser	{item:mediaContainer} IS NULL OR EXISTS ({{ SELECT 1 FROM {MediaContainer AS mc} WHERE {item:mediaContainer} = {mc.pk} }}

4.6 Critical Business Processes

Cronjobs are commonly used for triggering repeated tasks at a scheduled time. We have reviewed all your cronjobs and have found the following:

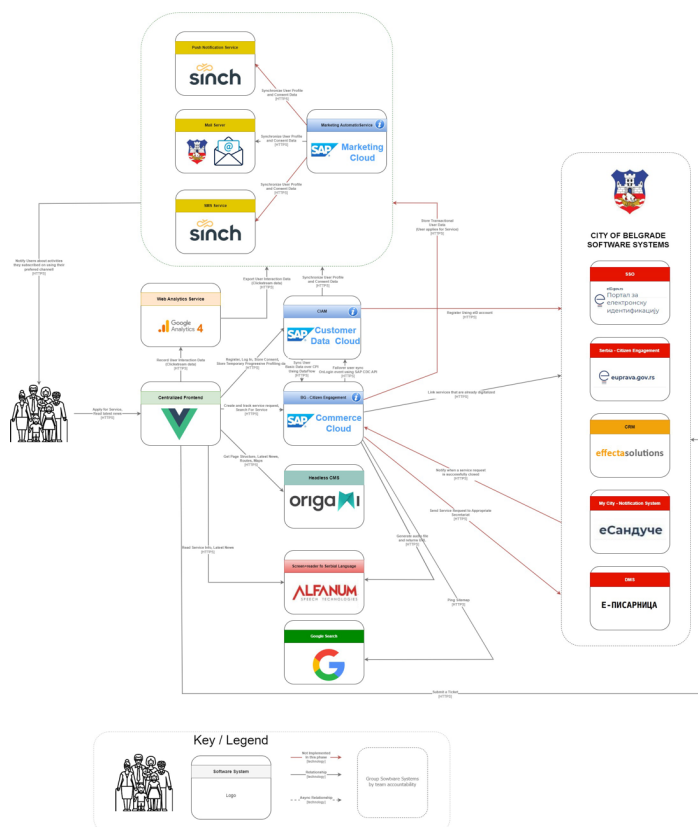
Cronjob	Recommendation
ALL	Not defined on the correct node. Recommendation is to set the nodeGroup to backgroundProcessing

5 INTEGRATIONS

5.1 High Level Architecture

Below is a high level architecture diagram used only for reference:

Belgrade Service 2 - System Landscape Diagram



5.2 Integration Practices

Although we are unable to complete a thorough review of your integrations we have completed an overall assessment and have outlined the following considerations:

Integration	Recommendation
-	All integrations (at the time of the review) follow the recommended best practices

6 TEST APPROACH

6.1 Test Data

It is recommended to have realistic test data for load testing your application.

At the time of the review 332 products were in the database. This seems a valuable amount for the purpose of the project.

6.2 Performance Testing

Performance testing is the most accurate way to predict how your production environment will handle your initial go-live and subsequent weeks.

Running performance tests is out of the scope of the Cloud Readiness Check, but we recommend you review [Managing Performance in an SAP Commerce Cloud Project](#) and become familiar with [Using Dynatrace to Manage and Optimize the Performance of your SAP Commerce Cloud Solution](#).

If you need detailed performance testing and tuning please see section 9 for Services that can assist with this task.

Performance tests are in place and were executed before the review.

Issue	Recommendation
Response time	Most of the pages have response times above 2 seconds. Some are above 3-4 seconds.

6.3 Environment Sizing

Although SAP Commerce Cloud provides CPU based [automated scaling](#) on Storefront and API aspects, it is important to ensure the size of your production environment is adequately sized for your baseline traffic, including your solr/backoffice/backgroundProcessing aspects.

Based our analysis of your setup and usage, it is recommended that your production environment be sized as follows:

Aspect	Sizing Recommendation
Storefront	As there is no storefront available the endpoint can be deleted and resources be moved to the API aspect.
API	Use storefront resources to increase performance
JS Storefront	As there is no storefront available the endpoint can be deleted and resources be moved to the API aspect.

Moving resources from one to another aspect can be done via ticket.

You should still run performance testing prior to go live with realistic data, including all the backgroundProcessing jobs to ensure the recommended size is valid.

7 DATA MAINTENANCE

Data maintenance is an essential part of an SAP Commerce Solution implementation, and one must consider the impact of the growing amounts of data that is generated on the overall health of the system. There needs to be a strategy to identify what data is live, what can be archived and what can be removed to keep the system running at peak performance and avoiding unplanned slowdowns months after the initial go-live.

The following part of this Cloud Readiness Check will review the implementation for the recommended practices identified in [Data Maintenance and Cleanup](#).

7.1 Large Tables

Based on our analysis the biggest tables are normal data tables.

Still keep in mind to implement a proper cleanup strategy even if there is no cleanup necessary at the moment.

Table	Rows count	Size (MB)	Recommendation
medias	161041	471.43	
solrindexoperation	286049	100.15	
attributedescriptorslp	424705	63.34	
attributedescriptors0lp	424705	61.63	
attributedescriptors	31146	21.78	
attributedescriptors0	31146	15.98	
composedtypes0lp	13592	2.73	
composedtypeslp	13592	2.59	
composedtypes	1350	1.93	
composedtypes0	1350	1.22	

7.2 CleanUp Cronjob

CleanupCronJobStrategy can be used to clean up instances of the CronJob. This strategy is based on maintenance framework created for processing instances of any type. For more information, see [cronjob service](#). Default settings for the cron job cleanup mechanism is to remove instances of the CronJob, which fulfil the following:

- Has no trigger
- Are successfully finished
- Are older than 14 days

SAP recommends to declare and activate all clean up cronjob necessary to keep database platform in good shape.

Beware: the information here can be interesting for an audit trail. Make sure to backup the database and only cleanup if there is no need for these entries anymore.

If it has been a long time since you have cleaned up your cronjob history you may consider doing a [one-time cleanup](#).

Check the appendix for a list of jobs that should be cleaned up.

7.3 Sample Data

Often SAP Commerce project is started by using sample data which can help developer to accelerate the customisation. At the end of the project, it is important to clean up this not relevant data in order to not disturb business team who will be in charge to maintain the system alive.

In the custom code there is still some code referencing sample data (electronics and apparel sample setup).

This should be reviewed and removed.

Make sure backoffice objects and business settings are relevant for the business

7.4 Update JSON

It is recommended to use an [Update JSON](#) to ensure that sample data will not be loaded into the system when a platform update is executed via Cloud Portal.

At the time of this review no update JSON configured.

7.5 Generic Audit

The feature 'Generic Audit' has been introduced in 6.6 in order fulfil the GDPR issued by the European Union. The auditing tracks every persistence action (CRUD) of the specified types in a change log and is enabled by default (currently disabled). The change logs are stored in a dedicated table per type in the database (table name: <deploymentName><numericTypeCode>sn), which might grow very quickly. Currently, generic auditing is enabled for the following types:

Audited Types	Recommendation
conditional.session-cart	Check if autiting is required
product	Check if autiting is required
paymentinfo	Check if autiting is required
paymentmode	Check if autiting is required
streamconfigurationcontainer	Check if autiting is required
customerreview	Check if autiting is required
cxsegment	Check if autiting is required
cartentry	Check if autiting is required
consenttemplate	Check if autiting is required
unit	Check if autiting is required
order	Check if autiting is required
integrationobjectitemattribute	Check if autiting is required
quoteentry	Check if autiting is required
city	Check if autiting is required
endpoint	Check if autiting is required
title	Check if autiting is required
cxusertosegment	Check if autiting is required
quote	Check if autiting is required
usergroup	Check if autiting is required

Audited Types	Recommendation
integrationobjectitemvirtualattribute	Check if autiting is required
userpasswordchangeaudit	Check if autiting is required
btgsegment	Check if autiting is required
sapb2bdocument	Check if autiting is required
csagentgroup	Check if autiting is required
outboundsyncstreamconfiguration	Check if autiting is required
saporder	Check if autiting is required
currency	Check if autiting is required
principalgrouprelation	Check if autiting is required
consent	Check if autiting is required
address	Check if autiting is required
region	Check if autiting is required
integrationobject	Check if autiting is required
b2bunit	Check if autiting is required
cart	Check if autiting is required
integrationobjectitemclassificationattribute	Check if autiting is required
user	Check if autiting is required
csticket	Check if autiting is required
destinationtarget	Check if autiting is required
basesite	Check if autiting is required
orderentry	Check if autiting is required
country	Check if autiting is required
language	Check if autiting is required
abstractcredential	Check if autiting is required
consumeddestination	Check if autiting is required
outboundsyncjob	Check if autiting is required
integrationobjectitem	Check if autiting is required
btgsegmentresult	Check if autiting is required
employee	Check if autiting is required
district	Check if autiting is required
abstractcontactinfo	Check if autiting is required
attributeddescriptor	Check if autiting is required
outboundchannelconfiguration	Check if autiting is required
partneraddress	Check if autiting is required
comment	Check if autiting is required

If you are not obliged to meet GDPR or other data protection regulations, disable this feature completely by setting the following property:

```
auditing.enabled=false
```

Otherwise, configure the auditing wisely by disabling logging for non critical types such as products (for more information see <https://help.sap.com/viewer/aa417173fe4a4ba5a473c93eb730a417/LATEST/en-US/3e2b8491449d43b391eb1ce9c73cbb7f.html>):

```
audit.product.enabled=false
```

Note: Configuring auditing via properties is possible on a type level only. If a finer level of granularity is required, use [Change Log Filtering](#).

SAP CX Field Services recommends to have a clean up strategy in place for types with configured generic audit. This can be done by either defining a clean up CronJob leveraging the provided interface (see [here](#)) or by a custom implementation.

7.6 Promotions

The promotions module of SAP Commerce Cloud internally uses a promotion engine that is entrusted to perform evaluations on cart and perform selected actions for eligible carts. The promotion engine is based on Drools engine. It is important to keep in mind that misuse of the promotion engine might cause performance issues.

The following articles will help you keep a clean set of promotions or to optimize your promotions for best performance:

- [Top 10 Recommendations for Improving the Performance of your Commerce Cloud Promotion Engine](#)
- [Maintenance Cron Job for Drools Rules](#)

7.7 Data Migration

During your go-live a data and media migration will likely be needed. Be it a copy from other SAP Commerce installation or initial load from other systems.

In case of migrations from an existing SAP Commerce installation. The following guidelines are available to assist with migrating database and media:

- [Migrating to SAP Commerce Cloud: Data and Media Migration](#)
- [Commerce Migration Toolkit](#)
- [Migrate to SAP Commerce Cloud: Migrate Media with AzCopy](#)

In case of initial load, or migration from other tools. The following can be leveraged:

- [Strategies for Loading Data in SAP Commerce Cloud](#)

7.8 Other Maintenance Tasks

The above is non-exhaustive, and as for any IT system always ensure that a collection and rolling system in place for all log and troubleshooting data, as well as automated database and filesystem backup and archival procedures.

8 APPLICATION SECURITY REVIEW

Access to features within Commerce Cloud should be restricted to the appropriate users and sensitive data should be stored properly. We have reviewed a number of common issues found with user access and have outlined only those affecting your solution in the following table (checks that passed are not included):

Issue	Recommendation	Issues Found
Default passwords prefilled	Ensure all default login/passwords are disabled by adding to a *.properties file: DO NOT DELETE the admin user. The password is rotated automatically	No
Transparent Attribute Encryption	Although the database is encrypted, you should consider not storing sensitive data (e.g. credit card number) in SAP Commerce Cloud and store it in the system where it is required. If it is required for SAP Commerce Cloud, ensure you are using Transparent Attribute Encryption (TAE) for these specific attributes, items: see documentation If you don't specify a custom encryption key the default one is used. Although that means that the values are encrypted, they are encrypted with a key that is included in each platform ZIP. So it can be considered as non-encrypted.	Yes
Password Encoding	Some passwords were found to not be encoded using the following flexible search <pre>SELECT {passwordencoding}, count(0) FROM {user} GROUP BY {passwordencoding}</pre> See documentation	Yes
Default Users Credentials	Users left with default password: Nothing found DO NOT DELETE the admin user. The password is rotated automatically in CCv2 and can be looked up on hcs_admin services. Sample users left with non-default password and enabled account: Nothing found Sample users left with non-default password and disabled account: BackofficeIntegrationAdministrator, BackofficeIntegrationAgent, cmseditor, cmsmanager, cmspublisher, cmsreviewer, cmstranslator, cmstranslator-Annette, cmstranslator-Seb, integrationmonitoringtestuser, integrationserVICetestuser, searchmanager, searchmanager These users are listed here for your information, but as disabled and having non-default password they are not a direct security risk.	No

9 FURTHER INFO

SAP is here to support you during and after your go-live. Here is a list of possible services included with your subscription that you may choose to use.

Deployment Readiness

If you require an additional review of your system around your go-live period we recommend you schedule a Deployment Readiness Continuous Quality Check from SAP Enterprise Support. Please find more details about this service [here](#).

Performance Health Check

The Performance Health Check service is comprised of periodic analysis of your SAP Commerce Cloud Production storefront to identify application hotspots and top website response time contributors. The aim of this service is to highlight areas of the SAP Commerce Cloud platform which may be operating with degraded service due to incorrect configuration or customization. Consult with your Customer Engagement Executive or Account Executive to determine how often you qualify for this service.

Value Maximization Assessment

The Value Maximization Assessment, which is available only to professional and enterprise customers, provides a one-time analysis of your individual adoption of SAP Commerce Cloud's native features and functionalities after you've gone live and compares them with recommended practices to provide recommendations for increasing the value realized via your digital commerce platform. This analysis will result in an assessment report on key areas of focus, an overall adoption scorecard, and recommendations to drive more adoption. The findings will be documented, delivered via real-time presentation, and result in a jointly defined ongoing adoption plan. Consult with your Customer Engagement Executive or Account Executive to determine how to use this service.

Go-Live Checklist

In addition, you may find some of the following articles helpful:

- [Checklist for Migrating to SAP Commerce Cloud in the Public Cloud](#)
- [Go-live Readiness for SAP Commerce Cloud](#)
- [Managing Performance in your SAP Commerce Cloud Project](#)
- [Using Dynatrace to Optimize the Performance of your SAP Commerce Cloud solution](#)
- [Implement Continuous Delivery with SAP Commerce Cloud](#)
- [Data Maintenance and Cleanup](#)
- [User Management](#)
- [Strategies for Loading Data into SAP Commerce Cloud](#)

10 APPENDIX

10.1 Cronjobs to be removed

Cronjob	Schedule	Status
0000002C-ImpEx-Import		FINISHED
0000002D-ImpEx-Import		FINISHED
00000054		FINISHED
0000005B		FINISHED
0000005C		FINISHED
0000005Q		FINISHED
0000005R		FINISHED
0000005S		FINISHED
0000005T		FINISHED
000001JN		FINISHED
full-bghB2CIndex-cronJob		FINISHED
ticketsStagnationJob		FINISHED

10.2 Sonar issues

No	Severity	Message	Location
1	BLOCKER	Add or update the header of this file.	hybris/bin/custom/bgh/bghpatches/src/rs/bgh/patches/release/BghDefaultPatch.java:
2	BLOCKER	A "NullPointerException" could be thrown; "orderData" is nullable here.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/OrderRefundEmailContext.java:45
3	BLOCKER	A "NullPointerException" could be thrown; "orderData" is nullable here.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/OrderNotificationEmailContext.java:40
4	BLOCKER	Add or update the header of this file.	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/validator/AbstractReviewValidator.java:
5	CRITICAL	Make "orderConverter" transient or serializable.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/ConsignmentCollectionReminderEmailContext.java:45
6	CRITICAL	Make "formatFactory" transient or serializable.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/ConsignmentCollectionReminderEmailContext.java:39
7	CRITICAL	Make "customerServiceUncollectedConsignmentStrategy" transient or serializable.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/ConsignmentCollectionReminderEmailContext.java:38
8	CRITICAL	Make "consignmentConverter" transient or serializable.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/ConsignmentCollectionReminderEmailContext.java:37
9	CRITICAL	Make "quoteConverter" transient or serializable.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/QuoteNotificationEmailContext.java:29
10	CRITICAL	Make "quoteService" transient or serializable.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/QuoteNotificationEmailContext.java:27
11	CRITICAL	Make "orderConverter" transient or serializable.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/OrderRefundEmailContext.java:27
12	CRITICAL	Make "consignmentConverter" transient or serializable.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/ReadyForPickupEmailContext.java:27

No	Severity	Message	Location
13	CRITICAL	Make "priceDataFactory" transient or serializable.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/NotPickedUpConsignmentCanceledEmailContext.java:33
14	CRITICAL	Make "consignmentConverter" transient or serializable.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/NotPickedUpConsignmentCanceledEmailContext.java:32
15	CRITICAL	Make "customerConverter" transient or serializable.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/CustomerEmailContext.java:25
16	CRITICAL	Make "priceDataFactory" transient or serializable.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/OrderPartiallyModifiedEmailContext.java:36
17	CRITICAL	Make "orderConverter" transient or serializable.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/OrderPartiallyModifiedEmailContext.java:35
18	CRITICAL	Make "orderEntryConverter" transient or serializable.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/OrderPartiallyModifiedEmailContext.java:34
19	CRITICAL	Make "orderConverter" transient or serializable.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/OrderCancelledEmailContext.java:26
20	CRITICAL	Make "consignmentConverter" transient or serializable.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/DeliverySentEmailContext.java:27
21	CRITICAL	Make "orderConverter" transient or serializable.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/OrderNotificationEmailContext.java:30
22	CRITICAL	Make "keyGenerator" transient or serializable.	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/service/impl/BghMediaServiceImpl.java:55
23	CRITICAL	Make "timeService" transient or serializable.	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/service/impl/BghMediaServiceImpl.java:54
24	CRITICAL	Make "bghCategoryDao" transient or serializable.	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/service/impl/BghCategoryServiceImpl.java:38

No	Severity	Message	Location
25	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghgigaoccaddon/src/rs/bgh/gigaoccaddon/constants/BghgigaoccaddonConstants.java:13
26	MAJOR	Add a nested comment explaining why this method is empty, throw an UnsupportedOperationException or complete the implementation.	hybris/bin/custom/bgh/bghintegration/src/rs/bgh/integration/setup/BghintegrationSystemSetup.java:23
27	MAJOR	Replace this use of System.out or System.err by a logger.	hybris/bin/custom/bgh/bghintegration/src/rs/bgh/integration/BghintegrationStandalone.java:45
28	MAJOR	Replace this use of System.out or System.err by a logger.	hybris/bin/custom/bgh/bghintegration/src/rs/bgh/integration/BghintegrationStandalone.java:44
29	MAJOR	Move the "orderConfirmationEmailProcess" string literal on the left side of this string comparison.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:716
30	MAJOR	Move the "orderConfirmationEmailProcess" string literal on the left side of this string comparison.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:696
31	MAJOR	Either re-interrupt this method or rethrow the "InterruptedException" that can be caught here.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:619
32	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:532
33	MAJOR	Either re-interrupt this method or rethrow the "InterruptedException" that can be caught here.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:627
34	MAJOR	This method has 95 lines, which is greater than the 75 lines authorized. Split it into smaller methods.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:515
35	MAJOR	The Cyclomatic Complexity of this method "execute" is 11 which is greater than 10 authorized.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:515
36	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:512
37	MAJOR	This method has 111 lines, which is greater than the 75 lines authorized. Split it into smaller methods.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:496

No	Severity	Message	Location
38	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:286
39	MAJOR	class "AcceleratorTestOrderData" has 55 methods, which is greater than the 35 authorized. Split it into smaller classes.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:63
40	MAJOR	Invoke method(s) only conditionally.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:732
41	MAJOR	Invoke method(s) only conditionally.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:726
42	MAJOR	Invoke method(s) only conditionally.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:720
43	MAJOR	Invoke method(s) only conditionally.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:701
44	MAJOR	Invoke method(s) only conditionally.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:679
45	MAJOR	Invoke method(s) only conditionally.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:669
46	MAJOR	Invoke method(s) only conditionally.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:658
47	MAJOR	Extract this nested try block into a separate method.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:615
48	MAJOR	Merge this if statement with the enclosing one.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:595
49	MAJOR	Split this class into smaller and more specialized ones to reduce its dependencies on other classes from 43 to the maximum authorized 40 or less.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:63
50	MAJOR	Refactor this code to not nest more than 3 if/for/while/switch/try statements.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:615

No	Severity	Message	Location
51	MAJOR	Refactor this code to not nest more than 3 if/for/while/switch/try statements.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:593
52	MAJOR	Refactor this code to not nest more than 3 if/for/while/switch/try statements.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/orders/AcceleratorTestOrderData.java:607
53	MAJOR	Refactor your code to get this URI from a customizable parameter.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/BghAfterTenantInitializedEventListener.java:52
54	MAJOR	Refactor your code to get this URI from a customizable parameter.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/BghAfterTenantInitializedEventListener.java:51
55	MAJOR	Refactor your code to get this URI from a customizable parameter.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/BghAfterTenantInitializedEventListener.java:50
56	MAJOR	Refactor your code to get this URI from a customizable parameter.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/BghAfterTenantInitializedEventListener.java:49
57	MAJOR	Refactor your code to get this URI from a customizable parameter.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/BghAfterTenantInitializedEventListener.java:48
58	MAJOR	Refactor your code to get this URI from a customizable parameter.	hybris/bin/custom/bgh/bghtest/src/rs/bgh/test/BghAfterTenantInitializedEventListener.java:47
59	MAJOR	Change the visibility of this constructor to "protected".	hybris/bin/custom/bgh/bghpatches/src/rs/bgh/patches/release/AbstractBghPatch.java:47
60	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghpatches/src/rs/bgh/patches/structure/BghCountryOrganisation.java:63
61	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghpatches/src/rs/bgh/patches/structure/BghShopOrganisation.java:108
62	MAJOR	Make sure the use of "this" doesn't expose partially-constructed instances of this class in multi-threaded environments.	hybris/bin/custom/bgh/bghpatches/src/rs/bgh/patches/structure/BghShopOrganisation.java:59
63	MAJOR	Use assertNull instead.	hybris/bin/custom/bgh/bghcore/testsrc/rs/bgh/core/integration/CheckoutWithExternalTaxesIntegrationTest.java:123

No	Severity	Message	Location
64	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcore/testsrc/rs/bgh/core/integration/CheckoutWithExternalTaxesIntegrationTest.java:119
65	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/ConsignmentCollectionReminderEmailContext.java:62
66	MAJOR	This class has 6 parents which is greater than 5 authorized.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/ConsignmentCollectionReminderEmailContext.java:33
67	MAJOR	This class has 6 parents which is greater than 5 authorized.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/QuoteNotificationEmailContext.java:25
68	MAJOR	This class has 7 parents which is greater than 5 authorized.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/ForgottenPasswordEmailContext.java:19
69	MAJOR	Update this method so that its implementation is not identical to "getOrder" on line 71.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/OrderRefundEmailContext.java:82
70	MAJOR	This class has 6 parents which is greater than 5 authorized.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/OrderRefundEmailContext.java:25
71	MAJOR	This class has 6 parents which is greater than 5 authorized.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/ReadyForPickupEmailContext.java:25
72	MAJOR	This class has 7 parents which is greater than 5 authorized.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/OrderPartiallyCanceledEmailContext.java:16
73	MAJOR	This class has 6 parents which is greater than 5 authorized.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/NotPickedUpConsignmentCanceledEmailContext.java:30
74	MAJOR	This class has 7 parents which is greater than 5 authorized.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/OrderPartiallyRefundedEmailContext.java:20
75	MAJOR	This class has 6 parents which is greater than 5 authorized.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/CustomerEmailContext.java:23

No	Severity	Message	Location
76	MAJOR	This class has 6 parents which is greater than 5 authorized.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/OrderPartiallyModifiedEmailContext.java:32
77	MAJOR	Update this method so that its implementation is not identical to "getOrder" on line 67.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/OrderCancelledEmailContext.java:78
78	MAJOR	This class has 6 parents which is greater than 5 authorized.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/OrderCancelledEmailContext.java:24
79	MAJOR	This class has 6 parents which is greater than 5 authorized.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/DeliverySentEmailContext.java:25
80	MAJOR	This class has 6 parents which is greater than 5 authorized.	hybris/bin/custom/bgh/bghfacades/src/rs/bgh/facades/process/email/context/OrderNotificationEmailContext.java:28
81	MAJOR	Refactor this method to reduce the number of assertions from 39 to less than 25.	hybris/bin/custom/bgh/bghcommerceweb/services/testsrc/rs/bgh/controller/BghProductControllerIntegrationTest.java:327
82	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcommerceweb/services/testsrc/rs/bgh/queues/impl/ProductExpressUpdateQueueTest.java:147
83	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcommerceweb/services/testsrc/rs/bgh/queues/impl/ProductExpressUpdateQueueTest.java:131
84	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcommerceweb/services/testsrc/rs/bgh/queues/impl/ProductExpressUpdateQueueTest.java:128
85	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcommerceweb/services/testsrc/rs/bgh/queues/impl/ProductExpressUpdateQueueTest.java:124
86	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcommerceweb/services/testsrc/rs/bgh/queues/impl/ProductExpressUpdateQueueTest.java:113
87	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcommerceweb/services/testsrc/rs/bgh/queues/impl/ProductExpressUpdateQueueTest.java:110

No	Severity	Message	Location
88	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcommercew ebservices/testsrc/rs/bgh/queues/impl/ ProductExpressUpdateQueueTest.jav a:101
89	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcommercew ebservices/testsrc/rs/bgh/queues/impl/ ProductExpressUpdateQueueTest.jav a:95
90	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcommercew ebservices/testsrc/rs/bgh/queues/impl/ ProductExpressUpdateQueueTest.jav a:92
91	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcommercew ebservices/testsrc/rs/bgh/queues/impl/ ProductExpressUpdateQueueTest.jav a:89
92	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcommercew ebservices/testsrc/rs/bgh/queues/impl/ OrderStatusUpdateQueueTest.java:12 3
93	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcommercew ebservices/testsrc/rs/bgh/queues/impl/ OrderStatusUpdateQueueTest.java:11 9
94	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcommercew ebservices/testsrc/rs/bgh/queues/impl/ OrderStatusUpdateQueueTest.java:10 8
95	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcommercew ebservices/testsrc/rs/bgh/queues/impl/ OrderStatusUpdateQueueTest.java:96
96	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcommercew ebservices/testsrc/rs/bgh/queues/impl/ OrderStatusUpdateQueueTest.java:91
97	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcommercew ebservices/testsrc/rs/bgh/queues/impl/ OrderStatusUpdateQueueTest.java:88
98	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcommercew ebservices/testsrc/rs/bgh/queues/impl/ OrderStatusUpdateQueueTest.java:85
99	MAJOR	Swap these 2 arguments so they are in the correct order: expected value, actual value.	hybris/bin/custom/bgh/bghcommercew ebservices/testsrc/rs/bgh/queues/impl/ OrderStatusUpdateQueueTest.java:10 5

No	Severity	Message	Location
100	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghcommercew ebservices/web/src/rs/bgh/context/impl /DefaultContextInformationLoader.java :334
101	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghcommercew ebservices/web/src/rs/bgh/context/impl /DefaultContextInformationLoader.java :329
102	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghcommercew ebservices/web/src/rs/bgh/context/impl /DefaultContextInformationLoader.java :312
103	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghcommercew ebservices/web/src/rs/bgh/context/impl /DefaultContextInformationLoader.java :266
104	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghcommercew ebservices/web/src/rs/bgh/context/impl /DefaultContextInformationLoader.java :234
105	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghcommercew ebservices/web/src/rs/bgh/context/impl /DefaultContextInformationLoader.java :173
106	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghcommercew ebservices/web/src/rs/bgh/context/impl /DefaultContextInformationLoader.java :111
107	MAJOR	Use the built-in formatting to construct this argument.	hybris/bin/custom/bgh/bghcommercew ebservices/web/src/rs/bgh/context/impl /DefaultContextInformationLoader.java :334
108	MAJOR	Use the built-in formatting to construct this argument.	hybris/bin/custom/bgh/bghcommercew ebservices/web/src/rs/bgh/context/impl /DefaultContextInformationLoader.java :329
109	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommercew ebservices/web/src/rs/bgh/xstream/Xm IXStreamMarshallerFactory.java:297
110	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghcommercew ebservices/web/src/rs/bgh/xstream/Xm IXStreamMarshallerFactory.java:277

No	Severity	Message	Location
111	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/xstream/XmIXStreamMarshallerFactory.java:262
112	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/xstream/XmIXStreamMarshallerFactory.java:249
113	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/auth/GuestAuthenticationToken.java:29
114	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/request/redirection/OAuthRedirectionFilter.java:118
115	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/mapping/converters/VoucherConverter.java:26
116	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/populator/HttpRequestPaymentInfoPopulator.java:65
117	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/populator/HttpRequestReviewDataPopulator.java:41
118	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/populator/HttpRequestCustomerDataPopulator.java:43
119	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/populator/HttpRequestUserSignUpDTOPopulator.java:38
120	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/conv/ConverterPostProcessor.java:33
121	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/conv/ConverterPostProcessor.java:27
122	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/v2/config/WebConfig.java:140

No	Severity	Message	Location
123	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/v2/helper/StoresHelper.java:50
124	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/v2/helper/StoresHelper.java:38
125	MAJOR	The Cyclomatic Complexity of this method "doFilterInternal" is 11 which is greater than 10 authorized.	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/v2/filter/UserMatchingFilter.java:52
126	MAJOR	Refactor your code to get this URI from a customizable parameter.	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/v2/filter/AbstractUrlMatchingFilter.java:22
127	MAJOR	Invoke method(s) only conditionally.	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/v2/controller/ProductsController.java:251
128	MAJOR	Invoke method(s) only conditionally.	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/v2/controller/ProductsController.java:192
129	MAJOR	Split this class into smaller and more specialized ones to reduce its dependencies on other classes from 54 to the maximum authorized 40 or less.	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/v2/controller/ProductsController.java:102
130	MAJOR	Assign this magic number 2 to a well-named constant, and use the constant instead.	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/v2/controller/ProductsController.java:541
131	MAJOR	Assign this magic number 2 to a well-named constant, and use the constant instead.	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/v2/controller/ProductsController.java:519
132	MAJOR	Invoke method(s) only conditionally.	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/v2/controller/ForgottenPasswordsController.java:57
133	MAJOR	Invoke method(s) only conditionally.	hybris/bin/custom/bgh/bghcommerceweb/services/web/src/rs/bgh/v2/controller/BaseController.java:84
134	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommerceweb/services/src/rs/bgh/customer/populator/ExtendedCustomerPopulator.java:39
135	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommerceweb/services/src/rs/bgh/stock/CommerceStockFacade.java:51

No	Severity	Message	Location
136	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommercew ebseervices/src/rs/bgh/stock/Commerce StockFacade.java:38
137	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommercew ebseervices/src/rs/bgh/stock/Commerce StockFacade.java:25
138	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommercew ebseervices/src/rs/bgh/stock/impl/Defau ltCommerceStockFacade.java:76
139	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommercew ebseervices/src/rs/bgh/stock/impl/Defau ltCommerceStockFacade.java:56
140	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommercew ebseervices/src/rs/bgh/stock/impl/Defau ltCommerceStockFacade.java:41
141	MAJOR	Complete the task associated to this TODO comment.	hybris/bin/custom/bgh/bghcommercew ebseervices/src/rs/bgh/stock/impl/Defau ltCommerceStockFacade.java:28
142	MAJOR	Complete the task associated to this TODO comment.	hybris/bin/custom/bgh/bghcommercew ebseervices/src/rs/bgh/stock/impl/Defau ltCommerceStockFacade.java:27
143	MAJOR	Complete the task associated to this TODO comment.	hybris/bin/custom/bgh/bghcommercew ebseervices/src/rs/bgh/stock/impl/Defau ltCommerceStockFacade.java:26
144	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghcommercew ebseervices/src/rs/bgh/cronjob/OldCart RemovalJob.java:44
145	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommercew ebseervices/src/rs/bgh/queues/util/Prod uctExpressUpdateElementPredicate.ja va:43
146	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommercew ebseervices/src/rs/bgh/queues/populato rs/OrderStatusUpdateElementPopulat or.java:21
147	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommercew ebseervices/src/rs/bgh/queues/impl/Abs tractUpdateQueue.java:113
148	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommercew ebseervices/src/rs/bgh/queues/impl/Abs tractUpdateQueue.java:23

No	Severity	Message	Location
149	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghcommerceweb services/src/rs/bgh/queues/channel/ OrderStatusUpdateChannelListener.ja va:28
150	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghcommerceweb services/src/rs/bgh/queues/channel/ ProductExpressUpdateChannelListene r.java:32
151	MAJOR	Is //NOSONAR used to exclude false-positive or to hide real quality flaw ?	hybris/bin/custom/bgh/bghcommerceweb services/src/rs/bgh/queues/populato rs/ProductExpressUpdateElementPop ulator.java:22
152	MAJOR	Refactor this method to reduce the number of assertions from 51 to less than 25.	hybris/bin/custom/bgh/bghcore/testsrc/ rs/bgh/core/job/CustomersWithOrders RetentionCronJobIntegrationTest.java: 81
153	MAJOR	Use assertNotNull instead.	hybris/bin/custom/bgh/bghcore/testsrc/ rs/bgh/core/job/TicketsRetentionCronJ obIntegrationTest.java:75
154	MAJOR	Refactor this method to reduce the number of assertions from 47 to less than 25.	hybris/bin/custom/bgh/bghcore/testsrc/ rs/bgh/core/job/OrdersRetentionCronJ obIntegrationTest.java:80
155	MAJOR	Add some tests to this class.	hybris/bin/custom/bgh/bghcore/testsrc/ rs/bgh/core/common/BghServiceLayer Test.java:39
156	MAJOR	Refactor this method to reduce the number of assertions from 49 to less than 25.	hybris/bin/custom/bgh/bghcore/testsrc/ rs/bgh/core/job/CustomersWithoutOrd ersRetentionCronJobIntegrationTest.ja va:79
157	MAJOR	"thread" is a "Thread".	hybris/bin/custom/bgh/bghcore/src/rs/b gh/core/job/AbstractPaginatedMultiThr eadJobPerformable.java:54
158	MAJOR	Assign this magic number 2 to a well-named constant, and use the constant instead.	hybris/bin/custom/bgh/bghcore/src/rs/b gh/core/search/solrfacetsearch/provide r/impl/BghCommerceCategoryFacetDi splayNameProvider.java:52
159	MAJOR	Assign this magic number 2 to a well-named constant, and use the constant instead.	hybris/bin/custom/bgh/bghcore/src/rs/b gh/core/search/solrfacetsearch/provide r/impl/BghCommerceCategoryFacetDi splayNameProvider.java:46
160	MAJOR	Update this logger to use "VolumeAwareProductPriceValueProvider.class".	hybris/bin/custom/bgh/bghcore/src/rs/b gh/core/search/solrfacetsearch/provide r/impl/VolumeAwareProductPriceValue Provider.java:45

No	Severity	Message	Location
161	MAJOR	Replace "Collections.EMPTY_SET" by "Collections.emptySet()".	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/search/solrfacetsearch/provider/impl/BghPsProductStepsResolver.java:48
162	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/event/SubmitOrderEventListener.java:109
163	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/event/SubmitOrderEventListener.java:101
164	MAJOR	Define and throw a dedicated exception instead of using a generic one.	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/actions/quote/AbstractQuoteDecisionAction.java:68
165	MAJOR	Update this logger to use "CreateBuyerOfferAction.class".	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/actions/quote/CreateBuyerOfferAction.java:27
166	MAJOR	Update this logger to use "CheckForQuoteAutoApprovalAction.class".	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/actions/quote/CheckForQuoteAutoApprovalAction.java:28
167	MAJOR	Update this logger to use "QuoteApprovalRejectedAction.class".	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/actions/quote/QuoteApprovalRejectedAction.java:26
168	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/service/impl/BghSetupImpexServiceImpl.java:47
169	MAJOR	Use the built-in formatting to construct this argument.	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/service/impl/BghSetupImpexServiceImpl.java:47
170	MAJOR	The Cyclomatic Complexity of this method "getContentFromSource" is 11 which is greater than 10 authorized.	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/function/impl/BghGetCategoryAddressFunction.java:66
171	MAJOR	Replace this character class by the character itself.	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/utills/BghStringUtils.java:50
172	MAJOR	Replace this character class by the character itself.	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/utills/BghStringUtils.java:50
173	MAJOR	Replace this character class by the character itself.	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/utills/BghStringUtils.java:50
174	MAJOR	Remove this unnecessary quantifier.	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/utills/BghStringUtils.java:38

No	Severity	Message	Location
175	MAJOR	Reduce the number of non-empty switch cases from 61 to at most 30.	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/utis/BghProductUtils.java:58
176	MAJOR	This method has 216 lines, which is greater than the 75 lines authorized. Split it into smaller methods.	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/utis/BghProductUtils.java:46
177	MAJOR	Format specifiers should be used instead of string concatenation.	hybris/bin/custom/bgh/bghcore/src/rs/bgh/core/retention/impl/DefaultCSTicketStagnationAction.java:53

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